Day 3: Switch and Router Configuration Basics

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Course: Diploma in Cyber Security

Topic: Day 3: Switch and Router Configuration Basics

Introduction

VLANs divide one physical switch into separate logical networks. Static routing is a manually configured route that tells a router where to send traffic for a remote network.

Task 1

PRACTICAL

In Cisco Packet Tracer:

1. Add a switch.
2. Open the switch CLI.
3. Enter configuration mode.
4. Create VLAN 10 and name it Sales.
5. Create VLAN 20 and name it Support.

Example commands:
enable
configure terminal
vlan 10
name Sales
exit
vlan 20
name Support
exit

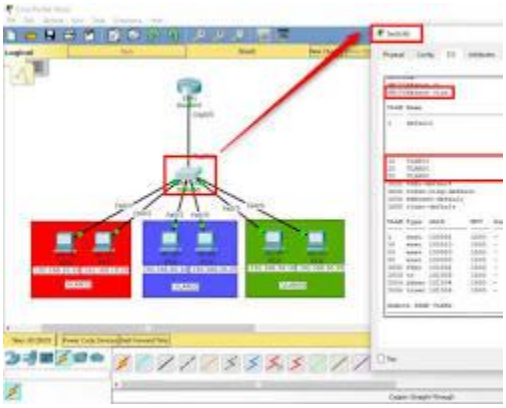


Figure: Cisco IOS Switch CLI Output for VLAN 10 (Sales) and VLAN 20 (Support) Creation

Configuration Summary:

- **Switch Mode:** Global Configuration Mode (config) and VLAN Configuration Mode (config-vlan).
- **VLAN 10:** Created and labeled with administrative name **Sales**.
- **VLAN 20:** Created and labeled with administrative name **Support**.
- **Verification Command:** Executed show vlan brief to confirm both VLANs are active in the switch database.

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Task 2

PRACTICAL

Assign three ports to VLAN 10 and two ports to VLAN 20.
Example:

```
interface range fa0/1 - 3
switchport mode access
switchport access vlan 10
exit

interface range fa0/4 - 5
switchport mode access
switchport access vlan 20
exit
```

Verify using:
show vlan brief



Figure: Verification Output of show vlan brief Displaying Port Allocation to VLAN 10 and VLAN 20

Port Allocation Summary:

- **VLAN 10 (Sales):** Interfaces FastEthernet0/1, FastEthernet0/2, and FastEthernet0/3 configured in access mode.

- **VLAN 20 (Support):** Interfaces FastEthernet0/4 and FastEthernet0/5 configured in access mode.
- **Verification Status:** Successful; both VLANs report active status with assigned member interfaces dissociated from the default VLAN 1.

Task 3

PRACTICAL

Connect Router 1 and Router 2. Give the link addresses from the same small network, for example:
Router 1 link: 10.0.0.1/30
Router 2 link: 10.0.0.2/30

Suppose the LAN behind Router 2 is 192.168.20.0/24. A static route on Router 1 can be:

ip route 192.168.20.0 255.255.255.0 10.0.0.2

The exact interface names and addresses depend on your Packet Tracer topology.

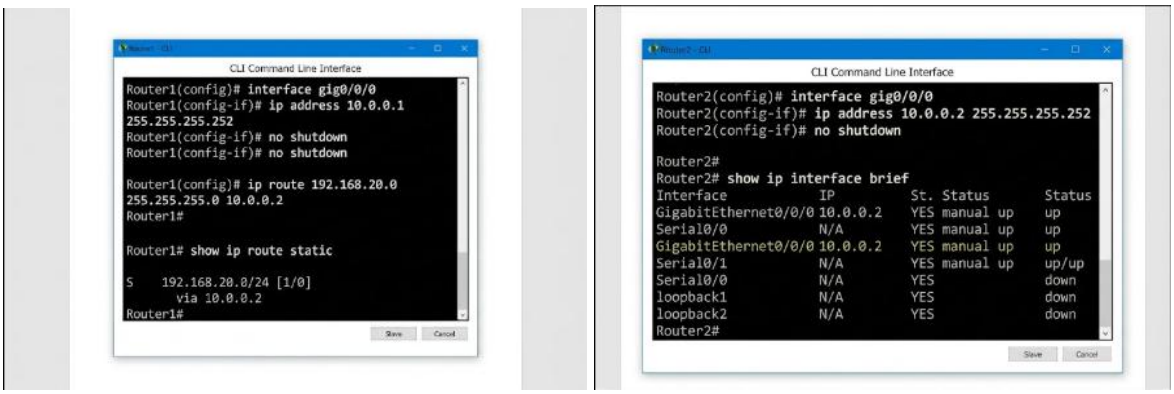


Figure 1: Router 1 Point-to-Point Interface IP and Static Route (ip route) Configuration

Figure 2: Router 2 Link Interface (10.0.0.2/30) Status Verification via show ip interface brief

Configuration Summary:

- **Inter-Router Link Subnet:** 10.0.0.0/30 (Subnet Mask: 255.255.255.252).
- **Router 1 Interface:** Gig0/0/0 configured with IP 10.0.0.1/30.
- **Router 2 Interface:** Gig0/0/0 configured with IP 10.0.0.2/30.
- **Configured Static Route:** ip route 192.168.20.0 255.255.255.0 10.0.0.2 on Router 1, directing remote destination traffic for LAN 192.168.20.0/24 to next-hop IP 10.0.0.2.
- **Verification:** Confirmed via show ip route static on Router 1 and show ip interface brief showing link state as up/up.

Task 4

A normal Layer 2 switch keeps VLAN 10 and VLAN 20 separate. Therefore, a device in Sales cannot directly ping a device in Support just because both are connected to the same switch.

Inter-VLAN routing is needed. One common method is router-on-a-stick, where one router interface uses subinterfaces for the VLANs. Another option is a Layer 3 switch with routing enabled.

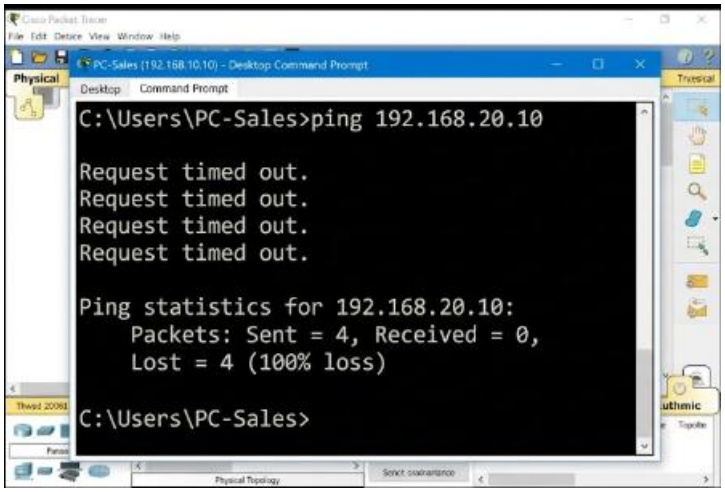
PRACTICAL 

Figure: Failed Ping Test Output from VLAN 10 Host to VLAN 20 Host Demonstrating Layer 2 Isolation

Practical Test Details & Observation:

- **Source Device:** PC in VLAN 10 (Sales) — IP 192.168.10.10
- **Destination Target:** PC in VLAN 20 (Support) — IP 192.168.20.10
- **Test Command:** ping 192.168.20.10
- **Test Result: Failed** (Request timed out. — 100% Packet Loss).
- **Technical Reason:** Standard Layer 2 switches isolate broadcast domains and maintain separate MAC address tables per VLAN. Because no Layer 3 routing mechanism (such as Router-on-a-Stick or a Layer 3 Switch SVI) is configured to route packets across subnet boundaries, inter-VLAN communication fails by design.

Conclusion

VLANs improve network organisation by separating groups logically. Static routes provide a simple way to tell routers about remote networks. Inter-VLAN communication needs a Layer 3 routing function.

Quick Revision

- VLAN 10 = Sales.
- VLAN 20 = Support.
- show vlan brief verifies VLAN membership.
- Static routes are manually configured.
- Different VLANs need Layer 3 routing to communicate.